

VAPO

Value-Model-Based Reinforcement Learning for Advanced Reasoning

60.4 on AIME 2024 (avg@32) — Surpassing DAPO by 10+ points

Base model: Qwen2.5-32B · No SFT data · ~5,000 training steps

ByteDance Seed · arXiv: 2504.05118

60.4

AIME 2024 Score

+10

vs. DAPO (50)

5K

Training Steps

Why Value Models? Finer Credit Assignment for Long CoT

Value-model-free methods assign trajectory-level advantage uniformly — inadequate for multi-step reasoning

VALUE-MODEL-FREE (GRPO, DAPO)

- Advantage from final trajectory reward only
- Uniform credit across all tokens
- Higher variance estimates (Monte Carlo)
- Subtle step-level errors go undetected

Limitation: cannot trace individual token impact



VALUE-MODEL-BASED (VAPO)

- Token-level credit assignment
- Traces each action's impact on returns
- Lower-variance per-token estimates
- Generalizes across samples → efficiency

Critical for long CoT where errors cascade

A well-trained value model generalizes across samples, improving sample efficiency — but training it for long-CoT introduces three core challenges.

Three Challenges Plaguing Value-Model-Based RL in Long-CoT

Why value models have underperformed despite theoretical advantages

1

Value Model Bias

BIAS

Long trajectories + bootstrapped learning accumulate systematic bias in value estimates, undermining policy gradients.

2

Heterogeneous Lengths

VARIANCE

Response lengths vary dramatically. Short and long sequences need different bias-variance tradeoffs in advantage estimation.

3

Reward Signal Sparsity

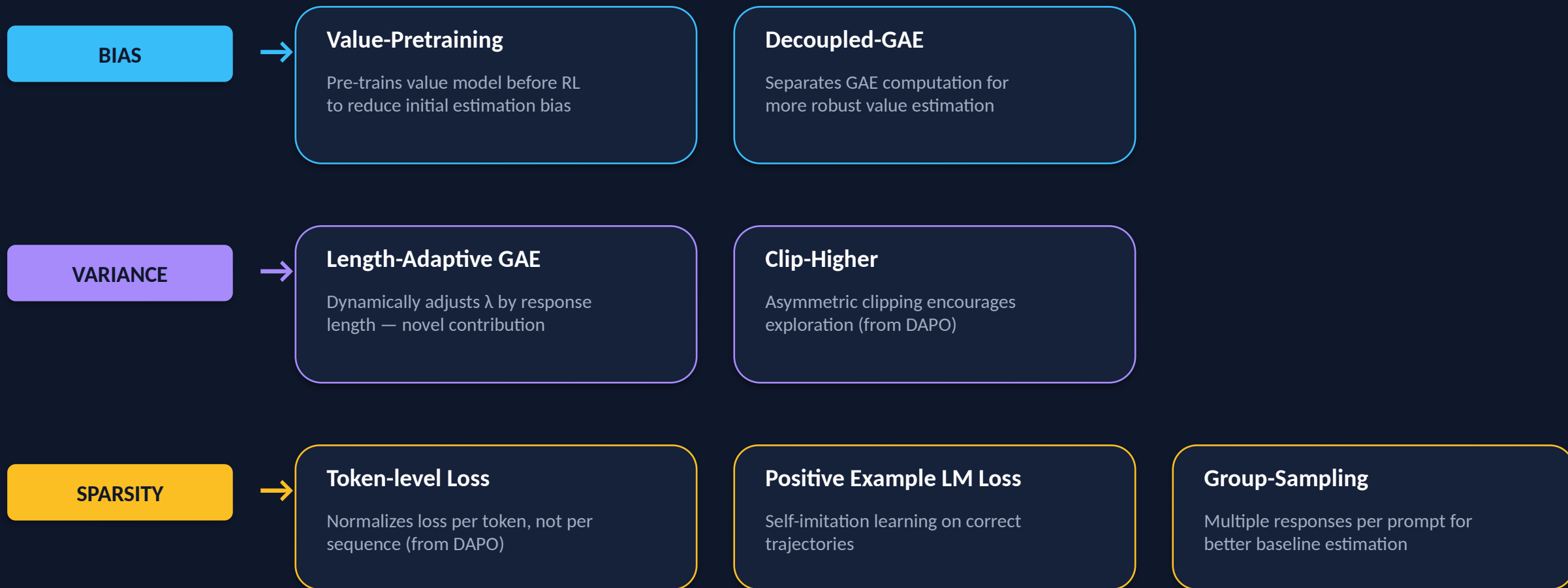
SPARSITY

Only the final <eos> token receives reward. Long CoT exacerbates signal dilution, making credit assignment extremely difficult.

VAPO addresses all three challenges with an integrated set of techniques →

VAPO Framework: Integrated Solutions for Value-Model-Based RL

Each component maps to a specific challenge in long-CoT value-model training



Components sourced from: DAPO (Clip-Higher, Token-level Loss), GRPO (Group-Sampling), VC-PPO (Decoupled-GAE). Novel: Length-Adaptive GAE, Value-Pretraining integration.

Length-Adaptive GAE: Dynamic λ for Variable Response Lengths

Novel contribution — optimizes bias-variance tradeoff across heterogeneous sequence lengths

SHORT RESPONSES

LOW λ

Lower $\lambda \rightarrow$ more reliance on value estimates

Value model bias is small for short trajectories

Prioritizes lower variance

Stable, tight advantage estimates



LONG RESPONSES

HIGH λ

Higher $\lambda \rightarrow$ more reliance on Monte Carlo returns

Value model bias grows with trajectory length

Prioritizes lower bias

Avoids compounding estimation errors

KEY INSIGHT

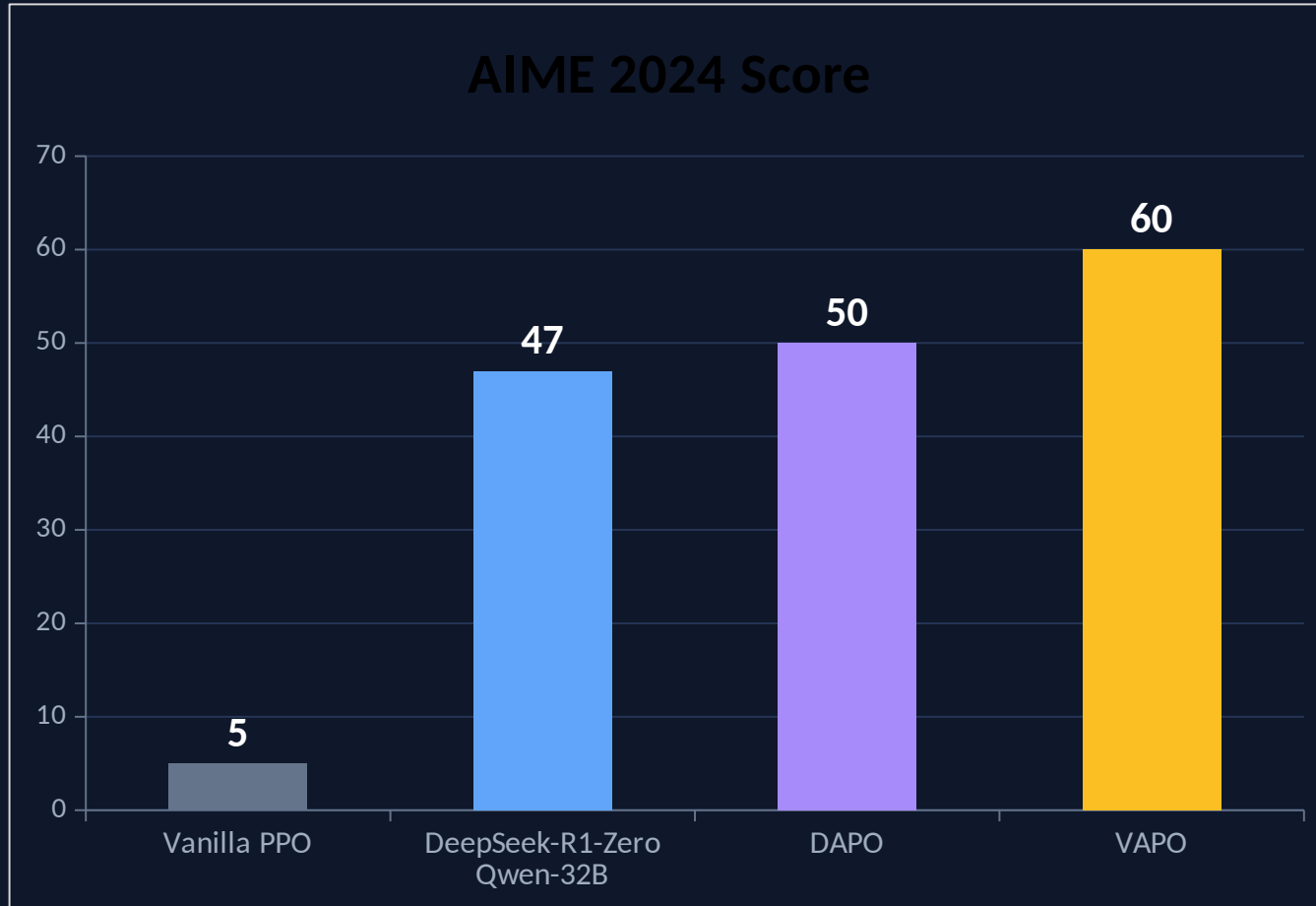
Standard GAE uses a fixed λ — a single tradeoff point for all sequences.

Length-Adaptive GAE recognizes that optimal λ depends on trajectory length, contributing +15 points to AIME score.

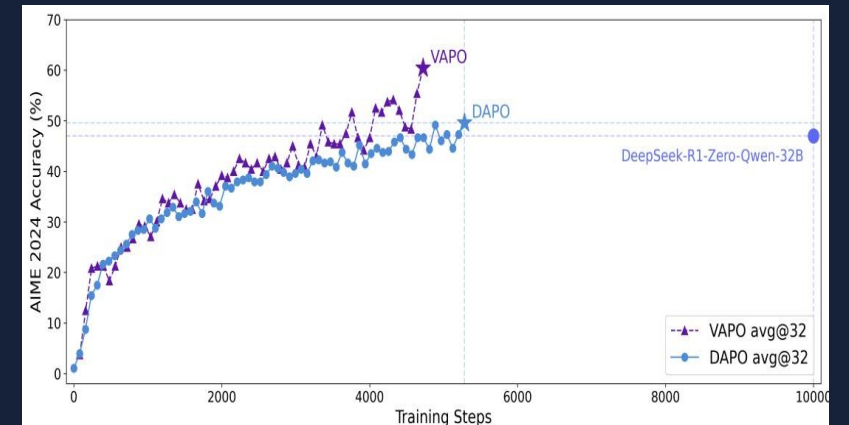
Ablation: removing Length-Adaptive GAE drops score from 60 $\rightarrow$ 45 (-15 points)

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO

All models trained on Qwen2.5-32B with no SFT data · AIME 2024 avg@32



TRAINING CURVE HIGHLIGHTS



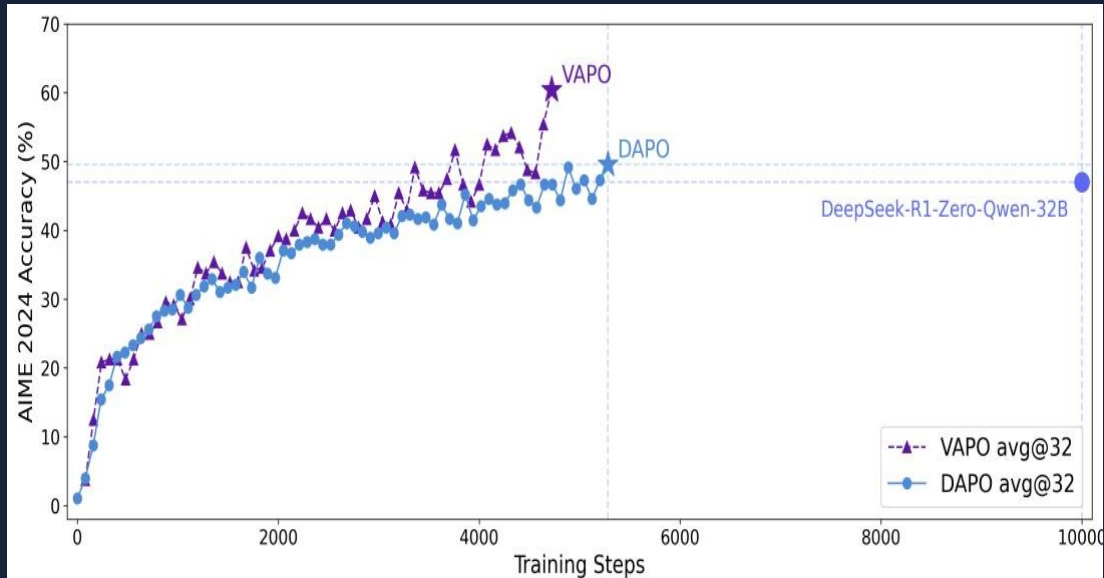
- VAPO reaches 60% at step ~4,500
- DAPO plateaus near 50% at step 5,500
- R1-Zero: 47% after ~10,000 steps

See figure_2.jpg for full training curve · VAPO consistently outperforms DAPO throughout training

Training Efficiency: SOTA in 5K Steps with Zero Crashes

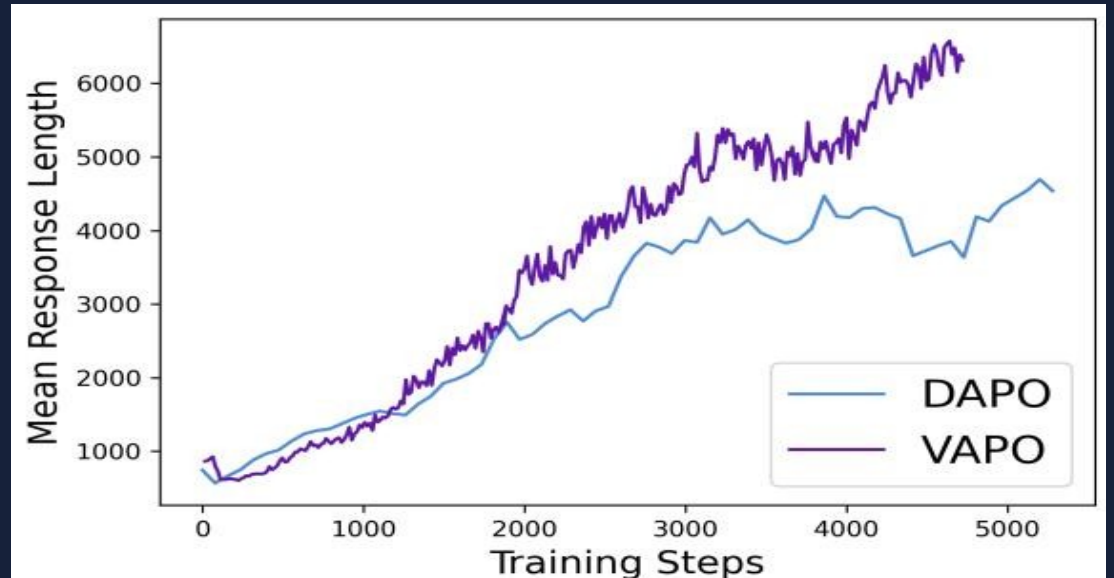
VAPO demonstrates superior convergence speed and training stability

AIME 2024 Accuracy vs Training Steps



- VAPO: 60% accuracy at ~4,500 steps
- DAPO: plateaus ~50% at 5,500 steps

Mean Response Length Over Training



- VAPO: ~6,000 tokens (longer reasoning chains)
- DAPO: ~4,500 tokens

~5,000 steps to SOTA
Training Budget

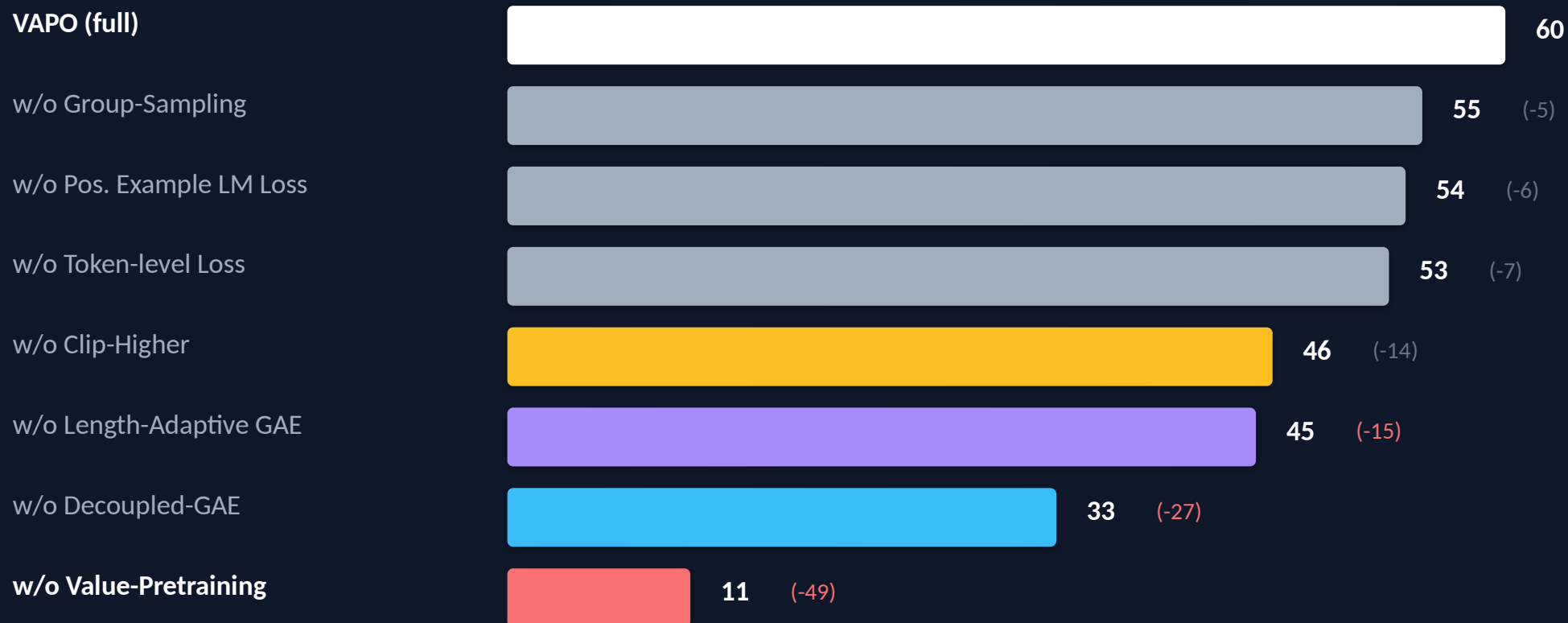
0 training crashes
Stability (multiple runs)

60% vs 50%
VAPO vs DAPO Final

~6,000 tokens
Avg Response Length

Ablation: Every Component Matters — Value-Pretraining Adds 49 Points

removing each component in isolation reveals its marginal contribution



CRITICAL: Removing Value-Pretraining collapses score from 60 → 11 (-49 points)

Without proper value model initialization, value-model-based RL barely functions. Decoupled-GAE (-27) and Length-Adaptive GAE (-15) are the next most impactful.

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1

Value-model-based RL surpasses value-free methods

VAPO (60.4) beats DAPO (50) and GRPO-based R1-Zero (47) on the same base model, proving value models work when the three core challenges are addressed.

2

Value-Pretraining is the single most critical component

Contributing 49 of the 60 points. Without it, value-model-based RL collapses to near-random (11).

3

Length-Adaptive GAE is a novel, high-impact technique

Dynamically adjusting λ by response length contributes +15 points, addressing the heterogeneous sequence length problem unique to long-CoT.

4

VAPO is stable and training-efficient

SOTA in ~5,000 steps with zero training crashes across multiple independent runs. DAPO needs ~5,500 steps for a lower score.

5

Implications for reasoning model development

The value-model paradigm is viable for complex reasoning. Proper initialization and length-aware advantage estimation are essential building blocks.